Axioms for physical reasoning: codifying the Seiberg–Witten solution in Lean

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Abstract

Most results in theoretical physics are not theorems. They are conclusions of arguments the community trusts without a proof from first principles. For the deepest of them a rigorous proof is conceivable but lies far out of reach—it would presuppose a nonperturbative construction of the interacting quantum field theory, a goal at least as hard as the Yang–Mills mass-gap Millennium problem. We argue that an interactive theorem prover is the right instrument for such a result even so: not to *prove* it—that is beyond reach—but to state the physical reasoning as a short list of **explicit, named axioms**, prove every mathematical consequence of them, and keep a machine-checked line between the two. The trust that remains then rests on that short, inspectable list, and the prover can report, for any downstream result, exactly which assumptions it used. We carry this out for the Seiberg–Witten solution of $\mathcal{N} = 2$ $SU(N)$ super-Yang–Mills, formalized in Lean 4. We describe what is proved, what is assumed, how the assumptions are checked—external review and an independent numerical oracle—and why this discipline is a sound standard for validating AI-generated results in theoretical physics. What we offer is a *discipline*, reviewable on its own terms: a reader may take the Seiberg–Witten mathematics on trust and still assess the formalization method.

1 Introduction

Most of what theoretical physics asserts is not proved. Across the field, the results we build on are typically not theorems but conclusions of *arguments*—chains of physical reasoning trusted because they cohere internally, because independent routes to the same quantity agree, and ultimately because they account for experiment. A result can be as certain as anything in science and still have no proof from first principles; certainty in physics rests on a different footing than in mathematics.

In quantum field theory the gap is at its widest. There the deepest results—the spectrum of a gauge theory, the phase structure of a model, an exact low-energy effective action—follow from reasoning about symmetry, holomorphy, dualities and consistency, and for the hardest of them a rigorous proof is not merely missing but out of reach: it would require first constructing the interacting quantum field theory nonperturbatively, a problem on the scale of the Yang–Mills mass-gap Millennium Prize. These results are as certain as anything in physics, and yet they are not theorems.

This gap has become pressing for a new reason: AI systems now generate physical derivations, and their characteristic error is not a wrong algebra step—a referee catches those—but a *hallucinated assumption*, an unphysical premise or a dropped subtlety presented as if derived. Catching that by reading is exactly what does not scale. What one wants instead is a way to make an argument's

assumptions explicit and machine-auditable, cleanly separating what is genuinely derived from what is taken on faith.

Interactive theorem provers are built for that separation, though physicists rarely meet them. A proof assistant such as **Lean 4** is a language in which one writes definitions and proofs that a small, fixed **kernel**—a few thousand lines of code, audited once and then trusted—checks line by line. A statement the kernel accepts is correct *relative to its inputs*: the handful of logical axioms built in, plus any further **axioms** the author explicitly declares. That whole list is the **trusted base**—everything one must believe in order to believe the result—and the prover can print it: for any theorem, the command `#print axioms` reports exactly which axioms it used, and nothing reaches that list without being named. Modern provers ship with large formal libraries of mathematics—Lean's is **Mathlib**, several hundred thousand theorems—so one formalizes atop real analysis, algebra and geometry rather than from scratch.

The point of this paper is to turn that machinery on a physics result that *cannot* be proved, and to use it honestly. We do not try to prove the Seiberg–Witten solution; that is beyond reach. Instead we **state the physical reasoning as a short list of explicit, named axioms, prove every mathematical consequence of them in Lean, and let the kernel hold the line between the two.** The residual trust is then concentrated on that named list; the kernel would expose a contradiction hidden among the axioms; and `#print axioms` audits, for any downstream claim, precisely which physical inputs it rests on. Justifying the axioms is still a physicist's task—here, external review and an independent numerical oracle—but it becomes a finite, inspectable one rather than a diffuse act of faith.

We take as the test case the **Seiberg–Witten solution** of N=2 SU(N) super-Yang–Mills: a celebrated, universally trusted result, unprovable within reach, yet unusually sharp in content—it identifies the exact low-energy physics with the period geometry of a specific family of algebraic curves, so both the physical assumptions and their mathematical consequences are crisp enough to formalize. Around it we obtain a clean split: a proved mathematical spine (the curve, its genus, positivity of the special-Kähler metric, the period matrix), the physical hypotheses as named predicates, and the headline statements—existence, and uniqueness up to duality—proved from those hypotheses plus a single mathematical input.

The rest of the paper proceeds as follows. §2 explains why a physics *result* is a different formalization target from a mathematical theorem, and §3 draws out the two kinds of formalization this entails. §4 states the Seiberg–Witten solution compactly. §5 is the formalization proper—what is assumed (the hypotheses, with their physical origins), what is proved, and two places where formalizing forced a hidden choice into the open. §6 describes how we check the axioms and §7 argues the discipline is the right standard for AI-generated physics. §8 lays out the remaining mathematical debt and where it will be discharged, and §9 the prospects for deriving the physical assumptions themselves from deeper principles. §10 places the work in context; §11 concludes. Two appendices give the Lean statements of the physical hypotheses (A) and the residual mathematical axioms (B).

2 Physics is mostly not theorems

A mathematician who formalizes a theorem has a proof to formalize. A physicist who formalizes a *result* usually does not. The Seiberg–Witten (SW) solution is the textbook case. No rigorous proof says that the low-energy physics on the Coulomb branch of $\mathcal{N} = 2$ $SU(N)$ gauge theory is governed by the curve $y^2 = P_N(x)^2 - \Lambda^{2N}$. One is *conceivable*—but it would first require constructing the interacting $\mathcal{N} = 2$ theory nonperturbatively, a problem at least as hard as the Yang–Mills mass-gap

Millennium problem, and so out of reach for the foreseeable future. What we have instead is an *argument*: a small set of physical assumptions—holomorphy from supersymmetry, the exact pattern of singularities, electric–magnetic duality, the weak-coupling asymptotics—that together pick out that curve, believed because three independent computations of the same quantity agree.

So "formalizing Seiberg–Witten" cannot mean "proving it"—not today, and not without first solving a Millennium-scale problem. That seems to put the result beyond formal methods. We argue the reverse: such a result is precisely where formalization helps most, *provided one is honest about what it does.*

> State the physical assumptions as **explicit, named axioms**; prove every **mathematical consequence** of them; and keep a machine-checked line between the two.

We call this second category—the non-rigorous reasoning physicists nonetheless trust—**foundations for physical reasoning**, by analogy with foundations of mathematics. A formalization then does three things no informal write-up can: it makes the dependency structure machine-checkable, it concentrates the residual trust onto a few named axioms, and it lets the kernel detect a contradiction among them.

This matters now because of *who*, or what, increasingly produces such arguments. As large language models begin to propose physical and mathematical derivations, their characteristic failure is not a wrong algebra step a referee would catch but a *hallucinated assumption*—an unphysical premise, or a dropped topological subtlety, presented as though derived. A machine-checked boundary between assumed and proved is precisely what keeps such a system honest; that is why we frame the discipline below as a standard for validating AI-generated physics, and return to it in §7.

3 Two kinds of formalization in physics

	What it is	**Examples**
Mathematical physics	A rigorous statement exists; you prove it.	Stability of matter; constructive QFT; the lattice-gauge mass gap—*and the proved spine of this project*: the curve, its genus, special-Kähler positivity, the period matrix.
Foundations for physical reasoning	Codifying the *non-rigorous* reasoning the community trusts, where no proof is within reach (a rigorous one would be Millennium-hard).	*The Seiberg–Witten argument itself.*

This is not a value judgment but a bookkeeping discipline, and it is necessary because the kernel does not enforce it. A theorem can be proved, axiom-free, and still be the **wrong statement**—a dropped hypothesis, a flipped sign, a domain quietly narrowed. And a physical assumption written as an innocent-looking definition can smuggle a falsehood past the kernel entirely. The discipline forces every physical input into the open as a named hypothesis, and every mathematical claim onto the proved side.

For a reader who formalizes *mathematics*, the delta is threefold:

- there is no proof to formalize—the result is unprovable within reach (§2), so the question is not "is the proof correct?" but "are the assumptions explicit and the derivations sound?";
- the physical lore becomes **named hypotheses**—predicates a theorem takes in its signature—never smuggled into a definition or a convenient default;
- the residual trust then localizes onto a short list, and `#print axioms` of any result audits exactly which assumptions it used.

Nothing here is special to physics-as-a-subject; it is what changes when the object you formalize is an *argument the community trusts* rather than *a theorem with a proof.*

4 The Seiberg–Witten solution in one paragraph

For a reader outside physics: the SW solution maps the exact low-energy behaviour of a quantum system to the geometry of a specific family of algebraic curves. What matters for the formalization is less the physical meaning of the terms below than the rigid mathematical data they impose.

On the Coulomb branch—coordinates $u_2, \dots, u_N$, rank $r = N - 1$—the low-energy theory is abelian $U(1)^r$ gauge theory. Its data are the special coordinates a_i and their duals $a_{D,i}$ (periods of a meromorphic differential), the coupling matrix τ_{ij}, and the central charge $Z = n_e \cdot a + n_m \cdot a_D$ of a BPS state, whose modulus is its mass. The SW claim is that all of this *is* the period geometry of $y^2 = P_N(x)^2 - \Lambda^{2N}$ (P_N a degree-N polynomial in a coordinate x, distinct from the moduli u). The assumptions that single out that curve are: holomorphy / non-renormalization (one holomorphic prepotential $\mathcal{F}$, one-loop-exact perturbation theory); the exact singularity structure (the branch is singular exactly where a BPS state goes massless, and nowhere else); electric–magnetic duality ($\mathrm{Sp}(2r, \mathbb{Z})$ monodromy); the $\mathbb{Z}_{2N}$ R-symmetry fixing the singularities; and matching to the one-loop-plus-instanton expansion at weak coupling.

5 The formalization

5.1 What is assumed

The physical content is six hypotheses **H0**–**H6**, each a named Lean **predicate or structure** (never a global axiom: a theorem that uses the physics carries their bundle `IsSWEffectiveTheory` as a *hypothesis*, so the assumptions live in its type, not the trusted base). They are **not independent postulates**: every one is a *specialization of* **N=2 supersymmetry**—the eight supercharges of the theory—together with the gauge data ($SU(N)$, asymptotic freedom) and unitarity. We give each with its physical origin and Lean carrier:

- **H0 — the carrier** (*N=2 SUSY + Coulomb-phase Higgsing*). A generic adjoint vev breaks $SU(N) \to U(1)^{N-1}$, leaving $r = N - 1$ abelian N=2 vector multiplets; their special coordinates a, a_D, coupling τ, and the charge lattice with its Dirac pairing are the data (`CoulombBase`/`Sheet`), on the universal cover of the moduli space minus its singularities.
- **H1 — special geometry** (*N=2 SUSY; unitarity*). Eight supercharges force the Coulomb branch to be rigid special Kähler: one holomorphic prepotential F with $a_D = \partial F/\partial a$, $\tau = \partial^2 F/\partial a^2$; $\mathrm{Im}\,\tau \succ 0$ is positivity of the gauge kinetic term, the no-ghost (unitarity) condition (`Sheet.SpecialGeometry`).

- **H2 — BPS singularities** (*the central-charge BPS bound*). The N=2 algebra carries a central charge Z; all states obey $M \geq |Z|$, saturated by BPS states. A singularity of the effective theory is a charged BPS state going massless, $Z_n \to 0$ (`Sheet.SingularitiesAreBPS`).
- **H3 — Picard–Lefschetz monodromy** (*running induced by a light charged state*). Encircling a point where the BPS state of charge n is massless, its one-loop logarithm produces exactly the symplectic Picard–Lefschetz transvection in n (`PicardLefschetzAtGenericStratum`).
- **H4 — electric–magnetic duality** (*Montonen–Olive / Seiberg–Witten duality*). The freedom in which charges count as "electric" is the symplectic group $\mathrm{Sp}(2r, \mathbb{Z})$ preserving the Dirac pairing and the central charge; charts glue by it (`Deck`; sheets form a duality-glued `SWAtlas`).
- **H5 — R-symmetry** (*the anomalous $U(1)_R$*). The classical $U(1)_R$ is broken by instantons to a discrete subgroup, acting on the moduli (order $2N$ for $SU(N)$) and permuting the singular locus (`RSymmetry`).
- **H6 — weak-coupling asymptotics** (*N=2 nonrenormalization + asymptotic freedom*). The prepotential is **one-loop exact** in perturbation theory; the remainder is a sum of instanton corrections weighted by the dynamical scale Λ^{2N} (`Sheet.MatchesAsymptotics`).

The rank-1 monodromy input `SameSWMonodromy` (Appendix B) is itself a consequence of H2+H3—which BPS state is light where fixes the monodromy. Full Lean definitions are in Appendix A, which also discusses factoring these as one general N=2 special-geometry structure that `IsSWEffectiveTheory` specializes.

5.2 What is proved

From the hypotheses of §5.1 (with the mathematical inputs of Appendix B), the following are **sorry-free theorems**, and `#print axioms` reports exactly what each rests on:

- **The headline — existence and uniqueness up to duality.** The Seiberg–Witten curve *realizes* such a theory (`sw_curve_admits_effective_theory`), and the hypotheses fix it **uniquely up to** $\mathrm{Sp}(2r, \mathbb{Z})$ **electric–magnetic duality** (`sw_effective_theory_unique_up_to_duality`; atlas-level `sw_unique_up_to_duality`)—uniqueness being a period-map rigidity statement. Each depends on the three logical axioms plus a *single* mathematical input, the period geometry `swPeriodLayer`; for $SU(2)$, `sw_su2_unique` is finer still (physics + two classical complex-analysis facts).
- the $SU(N)$ curve has **genus** $N - 1$;
- its period matrix τ is **symmetric with** $\mathrm{Im}\,\tau \succ 0$—positivity of the special-Kähler metric, the physical no-ghost condition;
- for $SU(2)$, the curve is singular at **exactly** $u = \pm\Lambda^2$—the monopole and dyon points, by a discriminant computation, not lore;
- the Picard–Lefschetz monodromy of a vanishing cycle is **symplectic**—why the monodromy group sits in $\mathrm{Sp}(2r, \mathbb{Z})$;
- the rank-1 modular function $\lambda = \theta_2^4/\theta_3^4$ (from Mathlib's Jacobi thetas) is a proved $\Gamma(2)$-**Hauptmodul**, with the cusp symmetry $1 - \lambda = \theta_4^4/\theta_3^4$.

The headline's machine-checked dependency, in full—logic plus one named period-geometry input, nothing else:

```
#print axioms sw_effective_theory_unique_up_to_duality
-- depends on axioms:
--   [propext, Classical.choice, Quot.sound,   -- logic
--    swPeriodLayer]                            -- the one mathematical input (period geometry)
```

The exact statements, the finer SU(2) footprint, and `swPeriodLayer` itself are in Appendix B.

5.3 Two things formalizing forced into the open

The type system can refuse a convenient falsehood. Our first attempt made the periods a, a_D and the coupling τ single-valued functions of the moduli. External review flagged this as *vacuous*: a global holomorphic map from the moduli space to the Siegel upper half-space is constant by Liouville's theorem—a free theory. The data must instead live on the universal cover of the moduli space minus its singular locus. The corrected types encode this, refusing the free-theory shortcut that prose slides over.

A surprise comes out derived, not assumed. Argyres–Douglas points—where the theory becomes an interacting CFT with no Lagrangian—look as though they need their own postulate. In the formalization they are instead a *definition*: the locus where two **mutually non-local** BPS charges (nonzero Dirac pairing) go massless together, built from the central charge and the charge lattice already in hand. (This locus is *empty* for pure $SU(2)$—the monopole and dyon go massless at the two different points $\pm\Lambda^2$—but the definition is general, correctly accommodating the matter flavours and higher-rank theories where AD points actually occur.) A phenomenon the theory should explain falls out as a consequence—a check the axioms pass rather than a cost they incur.

6 Checking the axioms: validation by independent agreement

Concentrating trust onto named axioms is worthwhile only if those axioms are then checked. We use three tools, cheapest first.

An independent numerical oracle. Where a statement has computable content, we evaluate it in a separate engine (mpmath / numpy) that shares no code or authorship with the Lean. A run of 44 checks at 40-digit precision confirms the assumed theta-function identities ($\theta_3^4 = \theta_2^4 + \theta_4^4$, $\theta_3 \neq 0$), verifies that the proved λ transformation laws are *faithful* (catching any sign or shift error), anchors the normalization ($\lambda(i) = \frac{1}{2}$), and checks the $SU(2)$ singularity locus. Tellingly, the oracle first caught a bug **in itself**—a fractional power evaluated on the wrong branch—before agreeing with the Lean to full precision: the method earning its keep. A numerical pass is necessary, not sufficient; it validates values, not logical structure (it cannot see a too-strong hypothesis or a vacuous one).

Cross-model and self-review against the pinned sources, recorded per result, separating "is it proved?" from "does it say what the source says?".

The physicist's triangle. The deepest reason the SW lore is trusted is that three independent routes to the prepotential agree: weak-coupling perturbation theory, the instanton expansion (Nekrasov's partition function, itself later made rigorous), and the exact periods of the curve. This

independent agreement is the empirical content behind the axioms, and our protocol attaches it as a cross-check rather than asserting it.

7 Why this is the right standard for AI-generated physics

When an AI produces a physics result, the dangerous failure is rarely a wrong algebra step a referee would catch. It is an **unstated assumption**—a *hallucination*, in machine-learning terms—slipped in as though derived. Reasoning from named physical axioms turns the kernel and type system into a guardrail that makes this impossible to hide:

- a physical input must be *named* to be used;
- every named input shows up when the prover lists the axioms behind a result;
- an honest definition—the universal-cover carrier, Argyres–Douglas as a derived locus—cannot encode a convenient falsehood, because the kernel checks the types;
- and *justifying* the axioms remains the physicist's job, done by the same independent-agreement checks that earn trust in the first place.

Formalization does not make the physics true. It makes the reasoning **auditable**—which is exactly what a result no one can prove from first principles needs.

8 What remains, and where it lives

The discipline also makes the remaining *mathematical* debt legible. For $SU(2)$ the uniqueness headline is, as just described, already a theorem modulo named classical-math axioms; the debt is to *discharge those axioms*, and it reduces to a few sharply-identified, Mathlib-scale developments: the Jacobi triple product (which discharges the theta identities and non-vanishing); the modular-λ covering $\mathbb{H}/\Gamma(2) \cong \mathbb{C} \setminus \{0, 1\}$ (whose invariance half is already proved, leaving the surjectivity); and the Gauss–Manin / variation-of-Hodge-structure layer over the moduli space. The covering rests in turn on the **Riemann mapping theorem** and the **Schwarz reflection principle**—general complex analysis, absent from Lean (RMT is formalized in Isabelle/HOL), which we have scoped as upstream contributions. The point is that "is this physics right?" becomes a precise, prioritized list of theorems to build.

9 Prospects: deriving the physical assumptions

The hypotheses of §5.1 are themselves *consequences* of the microscopic definition of N=2 super-Yang–Mills, reached by physical arguments of varying rigor. Some of them are, on inspection, **theorems**—and formalizing those would push the assumed frontier one level deeper, deriving the parent N=2 structure from inputs more primitive than its own data. Three tiers.

Already within reach. The **BPS bound** $M \geq |Z|$ is a positivity lemma about the centrally-extended N=2 algebra: a Hermitian combination of supercharges has a positive-semidefinite anticommutator matrix whose eigenvalues are $M \pm |Z|$, the BPS (short-multiplet) case being its kernel. This is finite-dimensional linear algebra—Mathlib's positive-semidefinite API suffices—and it derives the core of H2 from the algebra alone, with no quantum field theory. **Dirac–Schwinger–Zwanziger**

quantization splits into a theorem and a postulate: the electromagnetic field angular momentum of a dyon pair is the antisymmetric pairing $(q_1 g_2 - q_2 g_1)/4\pi$ (an explicit $\int \mathbf{x} \times (\mathbf{E} \times \mathbf{B})$), and requiring angular momentum to be quantized forces that pairing into $\mathbb{Z}$—the integer symplectic structure of the charge lattice (H0), with the duality group $\mathrm{Sp}(2r, \mathbb{Z})$ (H4) its automorphisms preserving the central charge. The computation is formalizable; the quantization is one further named postulate, in the spirit of the others.

Formalizable, but needing infrastructure. Special geometry (H1) follows from N=2 superspace invariance of the vector-multiplet action—formalizable once N=2 superspace is built; the one-loop exactness behind H6 rests on holomorphy and nonrenormalization (more tractable than the loop integral it constrains); the discrete R-symmetry (H5) is what the chiral anomaly leaves of $U(1)_R$, an index computation. Real, but substantial.

Beyond reach, for now. That N=2 SYM *exists* as an interacting four-dimensional theory, with these multiplets and a BPS spectrum carried by genuine states, presupposes constructing the quantum field theory—the same barrier as the solution itself. The algebra is exactly what one gets to *assume*; its realization is what cannot yet be assumed away.

The lesson mirrors the body. Just as the headline reduced to physics plus standard mathematics, the physics itself partly reduces: the BPS bound and the Dirac pairing are mathematics in disguise, and formalizing them would move the frontier from "assume special geometry, the BPS bound, and a symplectic lattice" to "assume the supersymmetry algebra and angular-momentum quantization." Each such step shrinks what must be taken on trust.

10 Related work

The assurance artifacts (correspondence index, fidelity review, numerical validation, axiom audit) follow the conventions of the author's own *formalization-assurance* project (`math-commons/formalization-assurance`), separate from and predating this work; the upstream mathematical contributions are framed for the Lean FRO's Tau Ceti library. The Riemann mapping theorem is formalized in Isabelle/HOL (Paulson); the Schwarz reflection principle and the SW solution appear unformalized before this work.

11 Conclusion

A rigorous proof of the Seiberg–Witten solution lies beyond reach—it presupposes constructing the quantum field theory itself—yet the result is as certain as anything in physics. A proof assistant can still treat it rigorously: by separating the machine-checked consequences from the physical lore, naming the lore as auditable axioms, and validating those axioms by independent agreement. Proved and assumed as separate, machine-checked columns; trust narrowed to a short vetted list; faithfulness enforced by types and by an independent oracle: this, we argue, is the right standard for validating AI-generated results in theoretical physics.

A The physics hypotheses (H0–H6) — predicates, not axioms

After the strengthening (§5.1), the physical inputs are **contentful predicates and structures**, not opaque axioms: they are *assumed as hypotheses* (bundled in `IsSWEffectiveTheory`) by the

results that use them, so they never enter the trusted base. The carrier itself *is* the data of an N=2 $U(1)^{N-1}$ EFT (H0). None of the following is an `axiom`.

One principle, specialized. The six are not independent postulates but facets of **N=2 supersymmetry** read off the Coulomb branch: SUSY fixes the special-Kähler prepotential (H1) and the central-charge BPS bound (H2); the BPS spectrum then dictates the monodromy (H3) and, with the symplectic charge lattice, electric–magnetic duality (H4); the chiral anomaly fixes the discrete R-symmetry (H5); and the N=2 nonrenormalization theorem fixes the weak-coupling form (H6). Only the *gauge data*—the group $SU(N)$, its one-loop β-function coefficient (asymptotic freedom), and the dynamical scale Λ—is added to single out this theory. A natural refactor, flagged as future work, is to formalize that parent **once**: an abstract N=2 special-geometry / BPS Coulomb branch (special coordinates, charge lattice with Dirac pairing, central charge with the BPS bound, anomaly-discrete R-symmetry, nonrenormalized prepotential), with `IsSWEffectiveTheory` obtained as its specialization at $SU(N)$, Λ. The hypotheses below would then be *derived projections* of one structure rather than seven siblings, and the matter and higher-rank theories would follow by changing only the gauge data.

```
-- H0  carrier = the data of an N=2 U(1)^{N-1} EFT
structure CoulombBase (r : ℕ) where
  Δ : Set (Fin r → ℂ); U : Set (Fin r → ℂ)
  hUopen : IsOpen U; hUsmooth : U = Δᶜ
structure Sheet (B : CoulombBase r) where
  V : Set (Fin r → ℂ); hVopen : IsOpen V; hVsub : V ⊆ B.U
  hVsc : SimplyConnectedSpace V
  a aD : (Fin r → ℂ) → (Fin r → ℂ)      -- special coordinates and duals (holomorphic)
  tau  : (Fin r → ℂ) → SiegelUpperHalfSpace r   -- coupling τ (Im > 0 typed)

-- H1  special geometry: a prepotential F with aD = ∂F/∂a, τ = ∂²F/∂a²
def Sheet.SpecialGeometry (s : Sheet B) : Prop :=
  ∃ F, ContDiffOn ℂ 2 F (s.a '' s.V) ∧
    (∀ u ∈ s.V, ∀ i, s.aD u i = partialDeriv F i (s.a u)) ∧
    (∀ u ∈ s.V, ∀ i j, (s.tau u).val i j = partialDeriv2 F i j (s.a u))

-- H2  massless on Δ as the limit Z_n → 0; massive off Δ
def Sheet.BecomesMasslessAt (s) (u₀) (n) : Prop :=
  Tendsto (fun u => s.centralCharge u n) (nhdsWithin s.V u₀) (nhds 0)
def Sheet.SingularitiesAreBPS (s) : Prop :=
  (∀ u ∈ s.V, ∀ n ≠ 0, s.centralCharge u n ≠ 0) ∧
  (∀ u₀ ∈ closure s.V ∩ B.Δ, ∃ n ≠ 0, s.BecomesMasslessAt u₀ n)

-- H3  Picard-Lefschetz: the monodromy IS an actual deck transformation acting as the
--     reflection in the SAME vanishing charge n as H2 (not the vacuous ∃ g, g = transvection n)
def PicardLefschetzAtGenericStratum (s : Sheet B) : Prop :=
  ∀ u₀ ∈ closure s.V ∩ B.Δ, ∀ n, n ≠ 0 → s.BecomesMasslessAt u₀ n →
    ∃ (s' : Sheet B) (D : Deck s s'), D.g = transvection n

-- H4  duality gluing: a ℤ-linear (Sp(2r,ℤ)) charge automorphism, covariant central charge
structure Deck (s s' : Sheet B) where
  g : ChargeLattice r ≃ₗ[ℤ] ChargeLattice r
  symplectic : ∀ n n', diracPairing (g n) (g n') = diracPairing n n'
  covariant  : ∀ u ∈ s.V ∩ s'.V, ∀ n, s'.centralCharge u (g n) = s.centralCharge u n

-- H5  R-symmetry: a bijection of EXACT order 2N (not id) fixing Δ
def RSymmetry (B : CoulombBase r) (N : ℕ) : Prop :=
  ∃ ω, Function.Bijective ω ∧ ω^[2*N] = id ∧
```

```
    (∀ k, 0 < k → k < 2*N → ω^[k] ≠ id) ∧ ω '' B.Δ = B.Δ

-- H6  one-loop over ROOTS (eigenvalue differences; nonzero for SU(2)) + instantonic tail
def Instantonic (Λ) (N) (F) : Prop :=     -- fixed classical part, Λ≠0, holomorphic coeffs
  Λ ≠ 0 ∧ ∃ c, (∀ k, Differentiable ℂ (c k)) ∧
    ∀ a, F a - classicalPrepotential a - oneLoopPrepotential Λ a
         = ∑' k, c (k+1) a * (Λ^(2*N))^(k+1)
def Sheet.MatchesAsymptotics (s) (Λ) (N) : Prop :=    -- H1's F + non-renormalized form
  ∃ F, (... the H1 relations for F ...) ∧ Instantonic Λ N F

-- the six, bundled: an N=2 U(1)^{N-1} Seiberg-Witten effective theory
structure IsSWEffectiveTheory (s : Sheet B) (Λ : ℂ) (N : ℕ) : Prop where
  specialGeometry : s.SpecialGeometry            -- H1
  singularities   : s.SingularitiesAreBPS        -- H2
  picardLefschetz : PicardLefschetzAtGenericStratum s   -- H3
  matching        : s.MatchesAsymptotics Λ N     -- H6
  rSymmetry       : RSymmetry B N                -- H5
```

`Deck` (H4) relates sheets and is proved to make the BPS mass descend to the base (`Deck.bpsMass_eq`). Sheets are organized into an `SWAtlas` (covering `B.U`, overlaps `Deck`-glued), so the duality gluing H4 is *part of the theory* and uniqueness is stated at the atlas level (`sw_unique_up_to_duality`). The one rank-1 *monodromy* input that remains a placeholder axiom is `SameSWMonodromy` (Appendix B): two candidate couplings f, g carry the same $\mathrm{Sp}(2,\mathbb{Z})$ monodromy around the punctures—physically fixed by which BPS state (monopole at $+\Lambda^2$, dyon at $-\Lambda^2$) goes massless where—the input to the rank-1 rigidity theorem `sw_su2_unique`.

B The residual mathematical axioms and the (now proved) headline

With the physics in predicates (Appendix A), the **only global axioms** beyond the three logical ones (`propext, Classical.choice, Quot.sound`) are *mathematical*. The headline implications are **proved theorems** (no `sorry`), each resting on the physical hypotheses plus a single named math axiom—`swPeriodLayer` (B.1); the finer covering/lift/θ axioms (B.2) are the genus-1 pieces of that period geometry, on which the rank-1 route runs.

B.1 — The headline theorems and the consolidated period axiom. The period-level (Picard–Fuchs / Gauss–Manin) geometry of the SW curve family is taken as one clean axiom; the headline follows, `#print axioms` = standard-3 + `swPeriodLayer` (no `sorry`):

```
-- the single consolidated period-level math axiom (the SW period geometry exists)
axiom swPeriodLayer (N : ℕ) (Λ : ℂ) (hN : 2 ≤ N) : SWPeriodLayer N Λ

-- HEADLINE (proved): uniqueness up to Sp(2(N-1),ℤ) duality, on a connected chart overlap
theorem sw_effective_theory_unique_up_to_duality (hN : 2 ≤ N) (s s' : Sheet B)
    (hconn : IsConnected (s.V ∩ s'.V))
    (h : IsSWEffectiveTheory s Λ N) (h' : IsSWEffectiveTheory s' Λ N) :
    Nonempty (Deck s s') :=
  sw_unique_of_swPeriodLayer (swPeriodLayer N Λ hN) s s' hconn h h'

-- HEADLINE (proved): existence — the SW curve (P monic, degree N, trace-free) realizes a theory
theorem sw_curve_admits_effective_theory (hN : 2 ≤ N) (P : Polynomial ℂ)
    (hPm : P.Monic) (hPdeg : P.natDegree = N) (hPtr : P.coeff (N - 1) = 0)
    (hsf : Squarefree (P ^ 2 - Polynomial.C (Λ ^ (2 * N)))) :
    ∃ (B : CoulombBase (N - 1)) (s : Sheet B), IsSWEffectiveTheory s Λ N ∧
```

```
      B.Δ = {u | ¬ Squarefree (swCurvePoly N Λ u)} ∧ swModulus N P ∈ s.V :=
  sw_exists_of_swPeriodLayer (swPeriodLayer N Λ hN) P hPm hPdeg hPtr hsf

-- PROVED (standard-3): the classical Coulomb branch has dimension N − 1 (the rank)
theorem coulombBranchDim (N : ℕ) : Module.finrank ℂ (Fin (N - 1) → ℂ) = N - 1
```

B.2 — Classical mathematics not yet in Mathlib — the only remaining global axioms, to discharge by porting (the covering is packaged as a structure):

```
structure ThricePuncturedUniformization where
  cover : UpperHalfPlane → {v : ℂ // v ∉ ({0, 1} : Set ℂ)}      -- the developing map λ
  gamma2_invariant : ∀ γ ∈ Gamma2, ∀ τ, cover (γ • τ) = cover τ
  surjective : Function.Surjective cover
  fiber : ∀ τ τ', cover τ = cover τ' ↔ ∃ γ ∈ Gamma2, γ • τ = τ' -- fibres = Γ(2)-orbits

-- the Γ(2) covering ℍ/Γ(2) ≅ ℂ\{0,1} exists   (discharge: Schwarz reflection ⇒ RMT ⇒ cover)
axiom AX_thrice_punctured_uniformization : Nonempty ThricePuncturedUniformization

-- lift uniqueness: same-monodromy developing maps agree up to a modular frame change
-- (discharge: classical; Forster, Ahlfors)
axiom AX_developing_map_rigidity
    {Λ : ℂ} {D : Set ℂ} {u₀ : ℂ} {f g : ℂ → ℂ}
    (U : ThricePuncturedUniformization) (hmono : SameSWMonodromy Λ f g) :
    ∃ m : ℂ → ℂ, AnalyticOnNhd ℂ (m ∘ g) D ∧ f =ᶠ[nhds u₀] (m ∘ g)

-- Jacobi quartic identity and  theta3 ≠ 0      (discharge: Jacobi triple product)
axiom AX_jacobi_quartic (τ : ℂ) : theta3 τ ^ 4 = theta2 τ ^ 4 + theta4 τ ^ 4
axiom AX_theta3_ne_zero (τ : UpperHalfPlane) : theta3 (τ : ℂ) ≠ 0

-- the family's local monodromy is the Picard-Lefschetz transvection
-- (discharge: Gauss-Manin / local-system layer over the moduli space)
axiom localMonodromy : ∀ {r : ℕ}, ChargeLattice r → (ChargeLattice r → ChargeLattice r)
axiom AX_picard_lefschetz_local {r : ℕ} (γ : ChargeLattice r) :
    localMonodromy γ = transvection γ
```

The machine's dependency printout for the rank-1 uniqueness theorem makes the separation literal—logic, classical-math debt, and the one physical input, in five lines:

```
#print axioms sw_su2_unique
-- 'sw_su2_unique' depends on axioms:
--   [propext, Classical.choice, Quot.sound,      -- logic
--    AX_thrice_punctured_uniformization,         -- classical math (Γ(2) covering)
--    AX_developing_map_rigidity,                  -- classical math (lift uniqueness)
--    SameSWMonodromy]                             -- physics (the SW monodromy)
```

The $SU(2)$ uniqueness theorem `sw_su2_unique` rests on exactly the two covering/lift axioms (`AX_thrice_punctured_uniformization`, `AX_developing_map_rigidity`) plus the physics `SameSWMonodromy`—and the three logical axioms (`propext`, `Classical.choice`, `Quot.sound`), which underlie everything and belong to neither list. (The physical refinement `H4'Duality` and its derived consequence `metric_posDef_of_h4'`—electric–magnetic duality keeps the special-Kähler metric positive along the whole monodromy orbit—are proved, standard-3.)

B.3 — The period-level layer (a structure; its existence is §B.1's single axiom). The monodromy input `SameSWMonodromy` is itself the conclusion of the *periods* of the SW curve—their Picard–Fuchs equation and Gauss–Manin monodromy. We package that layer as a structure and prove the B.1 goals modulo it, for SU(2) and for general SU(N):

```
-- the period-level debt, bundled as a *hypothesis* (so the theorems stay standard-3)
structure SWPeriodLayer (N : ℕ) (Λ : ℂ) where
  hN : 2 ≤ N                                              -- SU(N) needs rank N-1 ≥ 1
  rigidity : ∀ {B : CoulombBase (N-1)} (s s' : Sheet B), IsConnected (s.V ∩ s'.V) →
      IsSWEffectiveTheory s Λ N → IsSWEffectiveTheory s' Λ N → Nonempty (Deck s s')
  realize  : ∀ P : Polynomial ℂ, P.Monic → P.natDegree = N → P.coeff (N-1) = 0 →
      Squarefree (P^2 - Polynomial.C (Λ^(2*N))) → ∃ (B : CoulombBase (N-1)) (s : Sheet B),
        IsSWEffectiveTheory s Λ N ∧ B.Δ = {u | ¬ Squarefree (swCurvePoly N Λ u)} ∧ ...

-- the B.1 goals, now THEOREMS modulo the layer (all standard-3):
theorem sw_unique_of_swPeriodLayer (PL : SWPeriodLayer N Λ) ... : Nonempty (Deck s s')
theorem sw_exists_of_swPeriodLayer (PL : SWPeriodLayer N Λ) ... : ∃ B s, IsSWEffectiveTheory ...

-- rank-1 uniqueness routes through an AXIOM-FREE lemma (frame builds the gluing; covariance
-- proved from det = 1):
theorem su2_deck_of_periodFrame (s s' : Sheet B) (α β γ δ : ℤ) (hdet : α*δ - β*γ = 1)
    (hper : ...period frame on the overlap...) : Nonempty (Deck s s')  -- 3 logical axioms only
```

So "period frame ⇒ duality equivalence" is *proved*; only the period frame (the layer's `rigidity`) is assumed, as honest math debt. Taking the layer's *existence* as the single axiom `swPeriodLayer` (§B.1) discharges that hypothesis once, so the **named** headline theorems are closed, with `#print axioms` = standard-3 + `swPeriodLayer`. Discharging the layer—the period map $\oint \lambda_{SW}$, its Picard–Fuchs ODE and Gauss–Manin monodromy, the rigidity argument—is a few weeks at genus 1 (the elliptic case, where the λ/θ machinery already exists) and a Mathlib-scale variation-of-Hodge-structure project in general genus. A first version was caught (adversarial review) asserting a single globally-constant $SL(2,\mathbb{Z})$ frame across a possibly-disconnected overlap, and quantifying over the U(N) rather than SU(N) family; both are corrected (`IsConnected` overlaps; monic trace-free P).

The mathematical input, stated plainly. Collected, the headline's entire mathematical debt is *one* classical object—the **variation of Hodge structure of the Seiberg–Witten curve family**: the periods $\oint \lambda_{SW}$ varying holomorphically over the moduli u, with their Picard–Fuchs equation and $Sp(2r,\mathbb{Z})$ Gauss–Manin monodromy. The B.2 axioms are not independent inputs; they are exactly its **genus-1 classical pieces**—the $\Gamma(2)$ uniformization $\mathbb{H}/\Gamma(2) \cong \mathbb{C} \setminus \{0,1\}$ and developing-map rigidity (the rank-1 period map and its monodromy/rigidity), the Picard–Lefschetz transvection, and two Jacobi θ identities—and B.3 packages the same object as a structure at all ranks. The fiber side already exists: the SW curve embeds in `jacobian-challenge`'s hyperelliptic type with genus 1 proved (`swEllipticData_genus`, standard-3), and that library supplies the per-fiber periods, Riemann-bilinear positivity, Siegel/θ. The single capability that remains—at *every* rank—is the **variation**: that period integrals move holomorphically with a flat Gauss–Manin connection (Sp-monodromy), not yet formalized in Lean. The whole result reduces to that one capability: a few months at genus 1 (on the existing $\mathbb{H}$/modular/λ/θ machinery), a Mathlib-scale project for general SU(N).

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